DESSERTOPIA

PAYMENT TRANSACTION MANAGEMENT SYSTEM

Introduction :

The Payment Transaction Management module records and monitors payments for customer orders in the Dessertopia online dessert store. The PAYMENT table stores the payment ID, order ID, payment date, payment mode, payment status, and transaction amount.

Table Creation and Structure :

Create PAYMENT Table :

```
SQL> CREATE TABLE PAYMENT (
  2      PAYMENT_ID VARCHAR2(10) PRIMARY KEY,
  3      ORDER_ID VARCHAR2(10),
  4      PAYMENT_DATE DATE,
  5      PAYMENT_MODE VARCHAR2(20),
  6      PAYMENT_STATUS VARCHAR2(20),
  7      TRANSACTION_AMOUNT NUMBER(10,2),
  8      FOREIGN KEY (ORDER_ID) REFERENCES ORDERS(ORDER_ID)
  9  );

Table created.
```

Describe PAYMENT Table :

```
SQL> DESC PAYMENT;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 PAYMENT_ID                                NOT NULL VARCHAR2(10)
 ORDER_ID                                           VARCHAR2(10)
 PAYMENT_DATE                                       DATE
 PAYMENT_MODE                                       VARCHAR2(20)
 PAYMENT_STATUS                                     VARCHAR2(20)
 TRANSACTION_AMOUNT                                 NUMBER(10,2)
```

Insert Payment Record :

```
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY001', 'O001', DATE '2026-10-01', 'UPI', 'Success', 550);

1 row created.
```

```
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY002', 'O002', DATE '2026-10-02', 'Card', 'Success', 240);

1 row created.
```

```
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY003', 'O003', DATE '2026-10-03', 'Cash', 'Failed', 360);

1 row created.
```

Display Payment Records :

```
SQL> SELECT* FROM PAYMENT ORDER BY PAYMENT_ID;

PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY001     O001       01-OCT-26 UPI                  Success
               550

PAY002     O002       02-OCT-26 Card                 Success
               240

PAY003     O003       03-OCT-26 Cash                 Failed
               360


PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY004     O004       04-OCT-26 Net Banking          Pending
               300

PAY005     O005       05-OCT-26 UPI                  Success
               500

PAY006     O006       06-OCT-26 Card                 Failed
```

Successful Payment Transactions :

```
SQL> SELECT *
  2  FROM PAYMENT
  3  WHERE PAYMENT_STATUS = 'Success';

PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY001     O001       01-OCT-26 UPI                  Success
               550

PAY002     O002       02-OCT-26 Card                 Success
               240

PAY005     O005       05-OCT-26 UPI                  Success
               500


PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY007     O007       07-OCT-26 UPI                  Success
               400
```

Payment Mode CHECK Constraint :

```
SQL> ALTER TABLE PAYMENT
  2  ADD CONSTRAINT CK_PAYMENT_MODE
  3  CHECK (PAYMENT_MODE IN ('UPI', 'Card', 'Net Banking', 'Cash'));

Table altered.
```

Payment Status CHECK Constraint :

```
SQL> ALTER TABLE PAYMENT
  2  ADD CONSTRAINT CK_PAYMENT_STATUS
  3  CHECK (PAYMENT_STATUS IN ('Success', 'Failed', 'Pending'));

Table altered.
```

Transaction Amount CHECK Constraint :

```
SQL> ALTER TABLE PAYMENT
  2  ADD CONSTRAINT CK_PAYMENT_AMOUNT
  3  CHECK (TRANSACTION_AMOUNT >= 0);

Table altered.
```

Failed Payment Transactions :

```
SQL> SELECT *
  2  FROM PAYMENT
  3  WHERE PAYMENT_STATUS = 'Failed';

PAYMENT_ID ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY003     O003       03-OCT-26 Cash                 Failed
               360

PAY006     O006       06-OCT-26 Card                 Failed
               450
```

Transaction Count by Payment Mode :

```
SQL> SELECT COUNT(*) AS TOTAL_TRANSACTIONS
  2  FROM PAYMENT
  3  GROUP BY PAYMENT_MODE;

TOTAL_TRANSACTIONS
------------------
                 1
                 2
                 1
                 3
```

Payment Count by Status :

```
SQL> SELECT PAYMENT_STATUS, COUNT(*) AS TOTAL
  2  FROM PAYMENT
  3  GROUP BY PAYMENT_STATUS;

PAYMENT_STATUS            TOTAL
------------------- ----------
Failed                         2
Pending                        1
Success                        4
```

Card Transaction Count:

```
SQL> SELECT COUNT(*) CARD_TRANSACTIONS
  2  FROM PAYMENT
  3  WHERE PAYMENT_MODE ='CARD';

CARD_TRANSACTIONS
-----------------
                0
```